

GRAPH ANALYTICS – PART 2

Machine Learning & Graphs:

- machine learning project lifecycle – CRISP-DM (business understanding, data understanding, data preparation, modelling, evaluation, deployment)
- machine learning challenges
 - Data: Insufficient quantity of data, poor quality, non-representative data, irrelevant features
 - Issues: Managing big data, Designing a flexible schema, Developing efficient access patterns
 - Performance: Predictive accuracy, Training performance, Prediction performance
 - Storing the model (size)
 - Realtime
- The role of graphs in machine learning
 - Graph features :
 - Data management
 - Data analysis
 - Data visualization
 - The use of graphs in different ways in each of the four stages of a machine learning workflow:
 - Managing the data sources
 - Learning ,
 - Storing and accessing the model ,
 - Visualizing

Graph Data Engineering

- Big Data
 - Collect, Clean, Store, Access

- 4 (or more) V: Volume, Velocity, Variety, Veracity
- Use Case: tracking a suspect, use data from cellular tower, graph representation of a CTN, cluster view, state transition model
- Lambda Architecture:
 - traditional batch data pipeline + fast streaming pipeline for real time data + serving layer for responding to queries
 - indexing data in batch e speed layers
 - technology agnostic data processing architecture
 - latency, data consistency, scalability, fault tolerance, human fault tolerance,
- Use case - CTN scenario with batch and serving layers
- Graphs for Master data management (MDM)
 - Use case: fraud detection, graph for the transactions, spot the source of fraud
 - MDM vs Datawarehouse
 - Graph based MDM advantages: flexibility, extensibility, search capability, indexing capability
- Graphs data management: modelling, storage, processing
 - Sharding (splitting data horizontally across multiple servers)
 - Replication (copying data across multiple servers, for high availability and scalability purposes)
 - Native non native graph DB: performance issue during storage querying
 - Label property graphs
 - Use case: supply chain network, identify bottlenecks

Graph in machine learning applications

- The role of graphs:
 - Graph modelling /graph construction
 - Graph storage and learning process
 - Models storage
 - Graph visualization

- Managing data sources
 - Converting data to its graph representation
 - Graph modelling
 - Identify the data sources available for algorithm training purposes
 - Analyze the data available and evaluate the content, in terms of quality and quantity
 - Design the graph data MODEL
 - Define the data flow (known as the ETL process)
 - Import data into the graph
 - Perform post-import tasks
 - Collaborative filtering: User - item interaction matrix, Relevance function, Similarity between items, fast computation of similarity by using graphs
- Storing and accessing machine learning models
 - Use case: Monitoring a subject: Bayesian network, Dynamic Bayesian network, Contextual Markov chain
- The power of Data Visualization:
 - Choosing the right visualization

Graphs for Content-based recommendations

- Use case: Video rental store
- Three main components:
 - Item analyser
 - User profile builder
 - Recommendation engine
- Use case: User modelling on MovieLens dataset
 - Item modelling - Representing item features
 - User profile modelling – explicit and implicit rating
- Providing recommendation:

- 1 approach - connecting user profile to feature of interest – user-feature links
- 2 approach - Vector Space Model – projection of item and user profile into a common VSM, computes the similarity between user interests and items
- 3 approach - Recommend items similar to those the user liked in the past - the only option when not relevant representation of user profiles connected to features – user-item links (Dice similarity coefficient, KNN computation)
- *Hands-on-Sessions*

Graphs for Collaborative Filtering Recommendations

- Collaborative filtering adv and disadvantage - cold start problem
- Bipartite graph for the User-Item dataset – U projection, V projection – user and Item vector - sparse vector representation
- Compute nearest neighbours – compute similarity , order from highest, select top K, store
- Final graph model (bipartite+similarity among user, and among items)
- Providing recommendation: Clustering items, Segmenting users, Finding similar items
- Recommendation process: relevance score, recommendation
- Dealing with the cold start problem
- *Hands-on-Sessions*

Graphs applied in the Mobility Scenario

- Static and dynamic data
- Spazio-temporal series
- Geospatial data, property of Geo data
- Geospatial DB: Neo4j, Neo4J Spatial
- Road network: primal and dual graph
- Use case: Road network of Modena, centrality measures
- *Hands-on-Sessions*

Fraud Detection

- Definition of fraud – similar to anomaly detection
- Different fraud detection methods: Rule engine, decision tree, Graph model
- Fraud prevention VS Fraud detection
- Type of frauds
- The role of graphs:
 - fraudulent transactions
 - anomaly detection
 - vector representation
- Use case: finding the original point of credit card fraud
- Use case: Identifying a fraud ring
- Proximity based algorithms: Cluster-based, density-based, distance based
- *Hands-on-Sessions*

Graphs for text mining and NLP applications

- Common needs across applications - extensive use of text (Search Engines, Recommendation Systems, ChatBots, Vocal Assistants)
- Basic approach to predictive text tools: Splitting input into words, Storing word order, Generating next-word suggestions
- Limitations of basic approach: No normalization (e.g., stemming, lemmatization), No dependency parsing, Poor treatment of entities and stopwords
- Natural Language Processing core tasks:
 - Text classification
 - Sentiment analysis
 - Machine translation
 - Named entity recognition
 - Topic modeling
 - Text generation
 - Summarization

- Question answering
- Preprocessing steps: Tokenization, Stop Words Removal, Stemming and Lemmatization, Part-of-Speech (POS) Tagging
- NLP and Graphs:
 - Graph structure: Annotated text, Sentence, TagOccurrence, Token, Tag
 - Use of graphs to store NLP outputs: Preserves order and structure, Enables phrase pattern identification
 - Advantages of graph-based NLP: Naturally represents highly connected data, Enables extensions (e.g., sentiment analysis, similarity), Supports diverse use cases (e.g., word suggestions, QA)
- *Hands-on-Sessions*

Graph-based Retrieval-Augmented Generation (Graph RAG)

- Large Language Models (LLMs), LLM limitations: Lack of up-to-date information, Hallucinations, Limited adaptability
- Retrieval-Augmented Generation (RAG): Combines external retrieval with generation, Improves relevance and accuracy of LLM outputs
- RAG Architecture:
 - External Data
 - Embedding Model
 - Vector Database
 - Retriever
 - Generator
- RAG Benefits: Enhanced relevance and quality , Task versatility, Improved user trust and control
- RAG Issues: Ambiguity in unstructured context, Difficulty with complex relational queries
- Graph RAG:
 - Uses graph-structured data instead of plain text
 - Leverages knowledge graphs for improved structure and context

- Graph RAG Process:
 - Query Processing
 - Graph Traversal
 - Subgraph Retrieval
 - Information Integration
 - Response Generation
- Advantages / Challenges of Graph RAG
- Neo4j tools for Graph RAG
- *Hands-on-Sessions*

Knowledge Graphs

- Definition, use in practice: Search engines (Google), question-answering (Siri, Alexa), social networks (LinkedIn, Facebook).
- Key components: Concepts, classes, properties, relationships, and entity descriptions.
- Formal knowledge representations: RDF(S), OWL.
- Instance data, schema data, metadata.
- Knowledge graph vs. traditional graph:
 - Incorporates semantics, types, and hierarchies.
 - Not just data + graph, but data + knowledge + graph.
 - Explicit, declarative, annotated, non-hierarchical, large-scale.
- From the Web of Data to Knowledge Graphs
 - Web 1.0: Static pages connected by hyperlinks.
 - Web 2.0: Social collaboration, user-generated content.
 - Web 3.0: Semantic Web – uses ontologies, NLP, reasoning, AI.
 - Web 4.0: Real-time connection between all devices; intelligent, emotional Web.
- Semantic Web stack:
 - Syntax → Semantics → Context → Pragmatics → Experience.
 - Goals: well-defined meaning of information to enable human-machine

cooperation.

- DBpedia: Transform Wikipedia content into structured RDF triples.